

# JUNHOO LEE

Ph.D. Candidate, Seoul National University

Room 311, Building 18, Seoul National University, Gwanak-ro 1, Gwanak-gu, Seoul, Republic of Korea  
mrjunoo@snu.ac.kr | +82-10-4042-6255 | https://junhoo.me

## EDUCATION

### Seoul National University

*Ph.D. Candidate in Intelligence and Information*

Advisor: Prof. Nojun Kwak, Machine Intelligence & Pattern Analysis Laboratory (MIPAL)

Seoul, South Korea

Sep 2021 – Present (Exp. Aug 2026)

### Seoul National University

*B.Sc. in Electrical and Computer Engineering*

B.S. Thesis: *Improvement of Elevator Control Algorithm Using Reinforcement Learning*. Advisor: Sungroh Yoon

Seoul, South Korea

Mar 2017 – Sep 2021

## RESEARCH INTERESTS

My research studies how foundation models acquire reusable structure, and how that structure can be controlled, adapted, and diagnosed across language, vision, generative modeling, and embodied interfaces. My work on discrete diffusion language models introduces inference-time planning structure for controllable generation [C1], while my meta-learning papers study fast adaptation and task generalization through optimization-trajectory structure and flexible episodic interfaces [C6, C7]. I also work on model diagnosis, semantic robustness, and data-efficient transfer, including deep support vectors for pretrained-model interpretation [C5], black-box semantic fingerprinting for generative models [C2], multimodal temporal grounding [C4], and instruction-following failures in text-to-image generation [W1]. Recently, I have been extending this line toward vision-language-action and embodied foundation models, where pretrained semantic structure must remain reliable under changing instructions, visual observations, and action interfaces.

## PUBLICATIONS

- [C1] **Unlocking the Potential of Diffusion Language Models through Template Infilling**  
Junhoo Lee, Seungyeon Kim, Nojun Kwak  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.  
**Oral Presentation (485/10518=4.61%)**
- [C2] **CSF: Black-box Fingerprinting via Compositional Semantics for Text-to-Image Models**  
Junhoo Lee, Mijin Koo, Nojun Kwak  
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [C3] **Deep Edge Filter** (Co-first)  
Dongkwan Lee<sup>†</sup>, Junhoo Lee<sup>†</sup>, Nojun Kwak  
*The Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [C4] **What's Making That Sound Right Now? Video-centric Audio-Visual Localization**  
Hahyeon Choi, Junhoo Lee, Nojun Kwak  
*The IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [C5] **Deep Support Vectors**  
Junhoo Lee, Hyunho Lee, Kyomin Hwang, Nojun Kwak  
*The Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [C6] **Any-Way Meta Learning**  
Junhoo Lee, Yearim Kim, Hyunho Lee, Nojun Kwak  
*The AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
- [C7] **SHOT: Suppressing the Hessian along the Optimization Trajectory**  
Junhoo Lee, Jayeon Yoo, Nojun Kwak  
*The Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
- [W1] **Do Not Think About Pink Elephant!** (Co-first)  
Kyomin Hwang<sup>†</sup>, Suyoung Kim<sup>†</sup>, Junhoo Lee<sup>†</sup>, Nojun Kwak  
*CVPR Workshops – Responsible Generative AI*, 2024.
- [W2] **Coreset Selection for Object Detection**  
Hojun Lee, Suyoung Kim, Junhoo Lee, Jaeyoung Yoo, Nojun Kwak  
*CVPR Workshops – Dataset Distillation*, 2024.
- [W3] **Practical Dataset Distillation Based on Deep Support Vectors**  
Hyunho Lee, Junhoo Lee, Nojun Kwak  
*ECCV Workshops – Dataset Distillation Challenge*, 2024.

[J1]     **End-to-End Multi-Entity Customization**  
Wonhark Park\*, Jaehyun Lee\*, Wonsik Shin, **Junhoo Lee**, Nojun Kwak  
*IET Image Processing*, Accepted. 2026.

[J2]     **The Role of Teacher Calibration in Knowledge Distillation**  
Suyoung Kim, Seonguk Park, **Junhoo Lee**, Nojun Kwak  
*IEEE Access*, 2025.

AWARDS & HONORS

|                                                            |           |
|------------------------------------------------------------|-----------|
| BK21 Future Innovation Talent Bronze Prize (USD 1,000)     | 2023      |
| BK21 Outstanding Research Talent Fellowship (USD 3,500)    | 2023      |
| Yulchon AI Star Scholarship (USD 8,000)                    | 2022      |
| 3rd Place, SNU FastMRI Challenge (USD 3,000)               | 2021      |
| Kwanak Scholarship (full funded)                           | 2021      |
| National Science and Engineering Scholarship (full funded) | 2017–2018 |

RESEARCH PROJECTS

**Gradient-Descent-Based Meta-Learning Algorithms** (Samsung Electronics, Main Researcher, 2022–2023)  
**Self-Directed Visual Object Detection for Shopping Assistance** (IITP, Project Manager, 2022–2024)  
**AI Video Generation for Korean Traditional Media Art** (MCST/EASYWITH, Main Engineer, 2024–2026)  
**Physically Plausible Video Diffusion Modeling** (AI Star Fellowship/MSIT, Project Manager, 2025).

SKILLS

**Programming Languages:** Python, CUDA, R, C++, Verilog; **ML Development Stack:** PyTorch, TensorFlow, Slurm, Docker.

TALKS

**Compositional Semantic Fingerprinting (CSF) for Black-box Attribution of Text-to-Image Models**, SWCS 2026, Apr. 2026  
**Deep Support Vectors: Interpreting Deep Models via Support Vector Extraction**, KCC 2024, Jun. 26, 2024.

ACADEMIC SERVICE

**Conference Reviewer:** CVPR (2025, 2026), ICCV (2025), ECCV (2026), ICLR (2025, 2026), ICML (2026), NeurIPS (2024, 2025, 2026).

PATENTS

**Method and Apparatus for Edge Filtering to Improve the Robustness of Deep Learning Models**, Korean Patent 10-2025-0215458  
**Method and Apparatus for Controlling Unintended Object Generation in Image Generation Models through Prompt Reconfiguration**, Korean Patent 10-2025-0063788.